Distance-3 Surface-Code Memory on IQM Emerald

Internal Experimental Summary

DuQuantum IQM Challenge 2026 — 24 October 2026

Objective and main finding

We evaluated whether more frequent stabilizer extraction improves the storage of logical $|0_L\rangle$ (Z-memory) and $|+_L\rangle$ (X-memory) on IQM Emerald, while holding the *memory interval* approximately fixed at 15 μs. The measured implementation showed **no net benefit from additional intermediate rounds**: Z-memory became worse as the number of rounds increased, while X-memory remained close to a 50% logical error probability. Unencoded single-qubit memory performed better under the tested conditions.

Implementation and experimental design

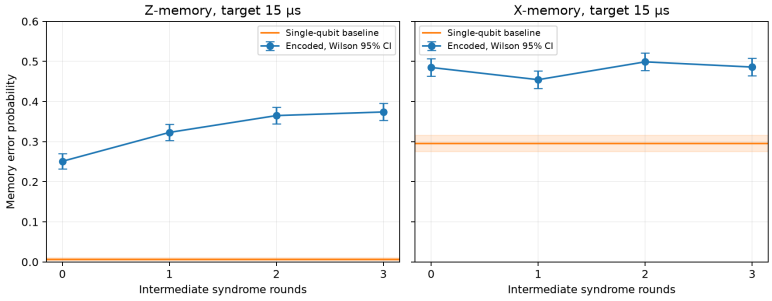
The experiment uses a rotated $[[9, 1, 3]]$ surface code with nine data qubits and eight ancillas on a fixed 17-qubit hardware patch. A common, unitary *known-state* preparation was used for each logical basis; it is not fault-tolerant preparation or an encoder for an arbitrary unknown input state. The qubit mapping was optimized and verified to require no additional routing SWAPs at the native-gate level. Intermediate ancilla resets were omitted. Measurement histories were converted to detection events, then decoded with PyMatching minimum-weight perfect matching (MWPM) using an arm-specific Stim detector error model. No physical correction gates were applied during storage. We also tried tuning decoder parameters, but the improvement was limited; alternative decoders remain an interesting direction to explore.

The 15 μs interval starts after preparation and ends before final basis rotation / destructive readout; it includes all intermediate extraction operations and idle delays. For $r = 0, 1, 2, 3$ rounds, available idle time was distributed approximately uniformly over $r + 1$ intervals. Locally compiled memory windows were 15.000 μs except for $r = 1, 2$ (14.996 μs). Timings were extracted from IQM pulse-compiler schedules rather than summed nominal gate lengths. Single-qubit $|0\rangle$ and $|+\rangle$ controls used the same defined memory interval. Each of 14 configurations was measured in two blocks of 1,000 shots, for **28,000 total shots**. All arms shared the saved calibration ID.

Measured results

15 μs memory configuration	Z error (95% CI)	X error (95% CI)
Encoded, 0 intermediate rounds	25.10% (23.25–27.05)	48.50% (46.31–50.69)
Encoded, 1 intermediate round	32.30% (30.29–34.38)	45.45% (43.28–47.64)
Encoded, 2 intermediate rounds	36.50% (34.42–38.63)	49.90% (47.71–52.09)
Encoded, 3 intermediate rounds	37.40% (35.31–39.54)	48.60% (46.41–50.79)
Unencoded single qubit	**0.65%** (0.38–1.11)	**29.55%** (27.59–31.59)

Table 1: Logical failure rates after decoding (encoded arms) and measured state error (unencoded controls); 2,000 shots per arm. Intervals are Wilson 95% binomial intervals, shown in percentage points.



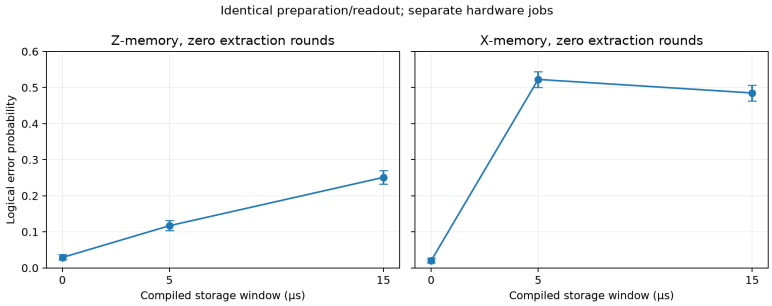
Zero-round storage diagnostic

To distinguish end-to-end preparation/readout error from storage-related degradation, we also ran zero-intermediate-round encoded memories at near-zero and 5 μs storage. The same complete state-preparation and final-readout procedures were

retained.

Encoded, zero intermediate rounds	Z error	X error
Near-zero memory interval	2.90%	2.00%
5 μs	11.70%	52.25%
15 μs	25.10%	48.50%

Table 2: Each entry represents 2,000 hardware shots. X-memory at 5 and 15 μs is close to random logical-parity prediction.



Component-control experiment (separate Z-memory run)

The three one-round controls were remeasured with shorter idle delays to match the historical three-round interval from the first complete syndrome record to the final-readout boundary: approximately 5.348 μs. The three-round measurement is reused unchanged. This is a separate Z-memory comparison, not the fixed-15 μs benchmark; the pipelined three-round circuit has no verified common independent encoding boundary. The acquisitions occurred in different jobs, so drift remains possible. The measured three-round-minus-idle LER difference is +18.60 percentage points (95% difference interval +16.03

Circuit	Decoded LER (95% CI)	CZ count	Shots
1 round + idle	13.95% (12.50–15.54)	19	2,000
1 round + CZ echo	14.25% (12.79–15.85)	67	2,000
1 round + ancilla measurements	12.50% (11.12–14.02)	19	2,000
3 complete rounds	**32.55%** (30.53–34.64)	67	2,000

Table 3: Remeasured one-round controls and the historical three-round reference; 2,000 shots per arm. Extra CZ gates were paired echoes; auxiliary ancilla measurements were not included in the decoder.

to +21.14). These controls do not uniquely identify a physical noise mechanism. The CZ-echo and repeated-ancilla-readout differences were +0.30 and -1.45 percentage points, respectively. Neither contrast was statistically detected after Holm correction over the three comparisons.

Discussion and takeaway

At fixed 15 μs storage, added extraction rounds increased Z-memory LER but did not improve X-memory. In the updated record-window-matched comparison, the measured three-round-minus-idle LER difference is +18.60 percentage points (95% difference interval +16.03 to +21.14). These controls do not uniquely identify a physical noise mechanism. Near-zero storage errors of only 2–3% point to substantial memory-interval degradation. Overall, the tested encoded memories did not beat unencoded qubits. Zero-SWAP mapping was verified and decoder parameter tuning gave limited improvements; other decoders and noise models remain worth exploring.

Source: provided `submission(1).zip`, especially `METHODS.md`, `RESULTS.md`, `validation.json`, and archived measurement tables. Figures are reproduced from the submission package.